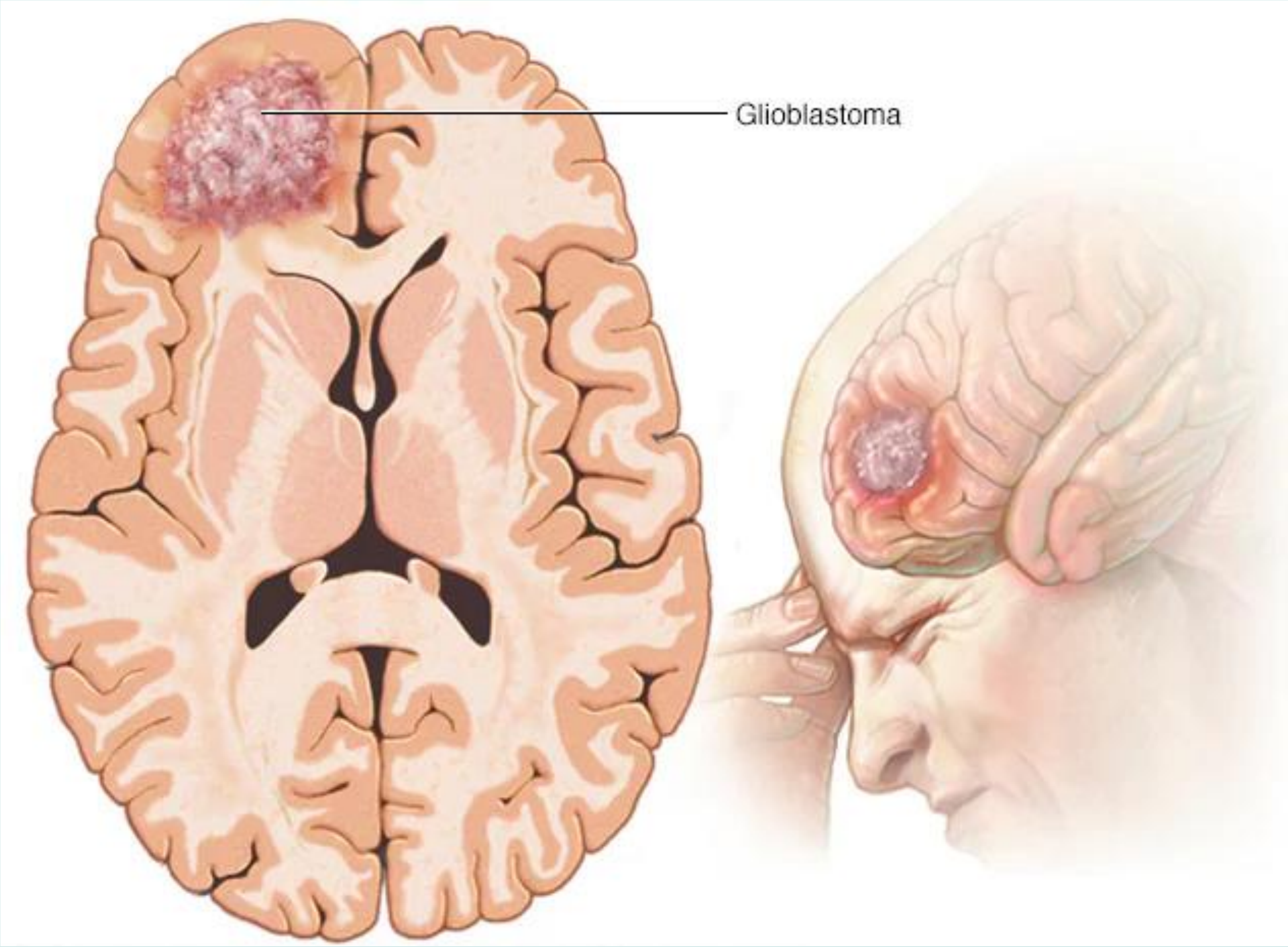


Mayo Foundation for Medical Education and Research

Mubritinib & Glioblastoma

Investigating a potential therapeutic treatment for a deadly and treatment-resistant cancer.

Caleb Ji – BCB420 – March 17, 2026

Motivation

Glioblastoma difficulties:

Intra-tumoral heterogeneity by the brain tumor stem cells (BTSCs).[1]

Multiple metabolic dependencies causing therapeutic resistance.[2]

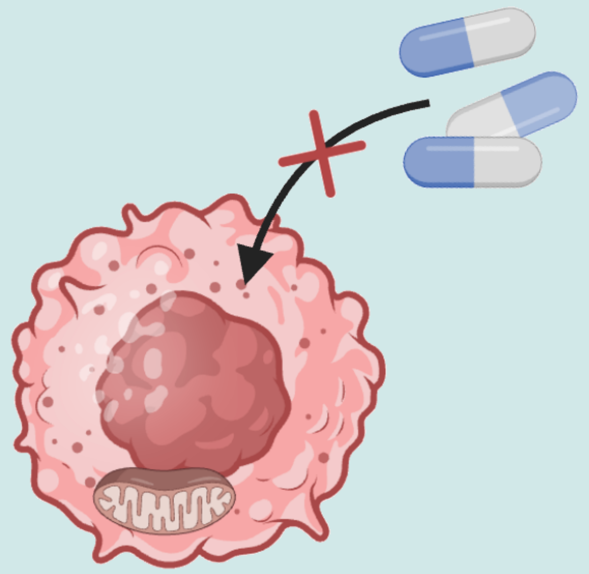


Figure created with biorender:

1. Prager, 2020
2. Agnihotri and Zadeh, 2016

Motivation

Glioblastoma difficulties:

Intra-tumoral heterogeneity by the brain tumor stem cells (BTSCs).[1]

Multiple metabolic dependencies causing therapeutic resistance.[2]

Mubritinib possibilities:

Shown to inhibit mitochondrial oxidative phosphorylation in amyloid leukemia.[3]

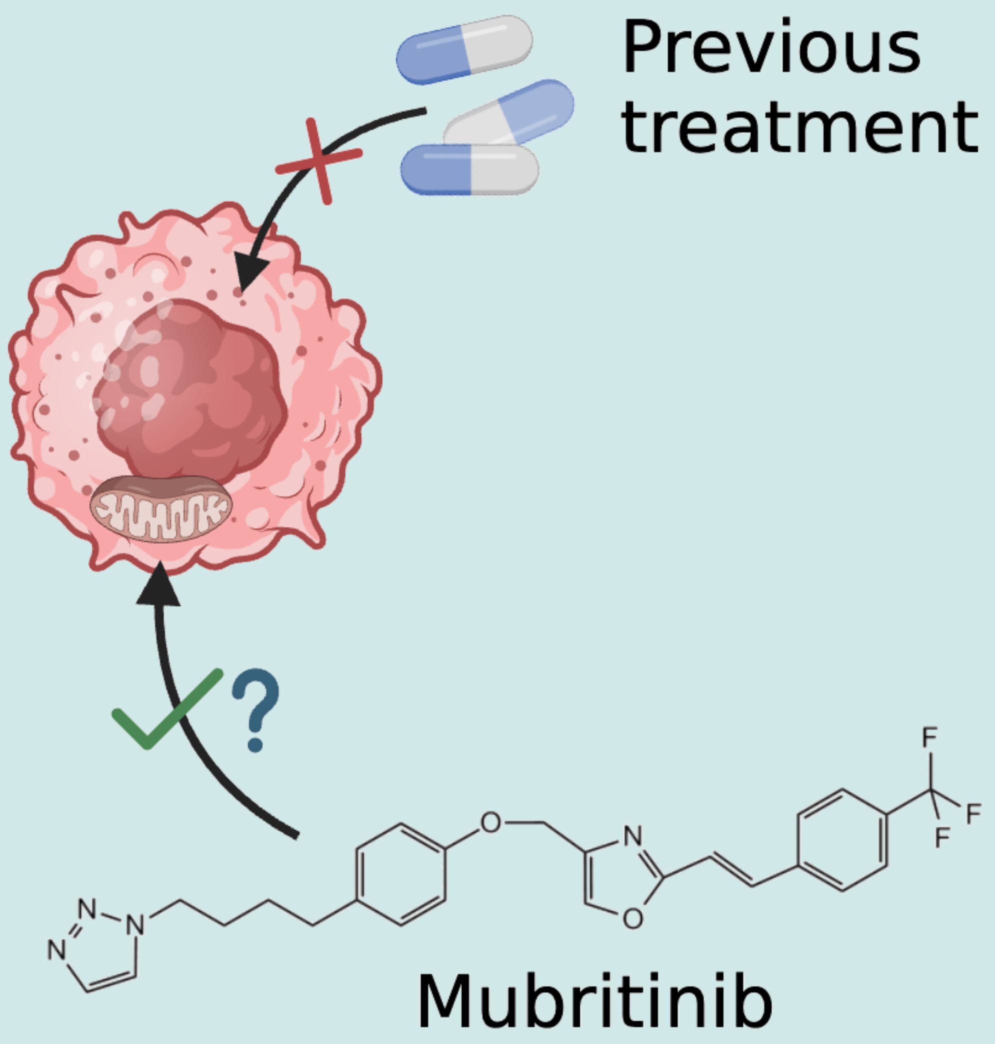
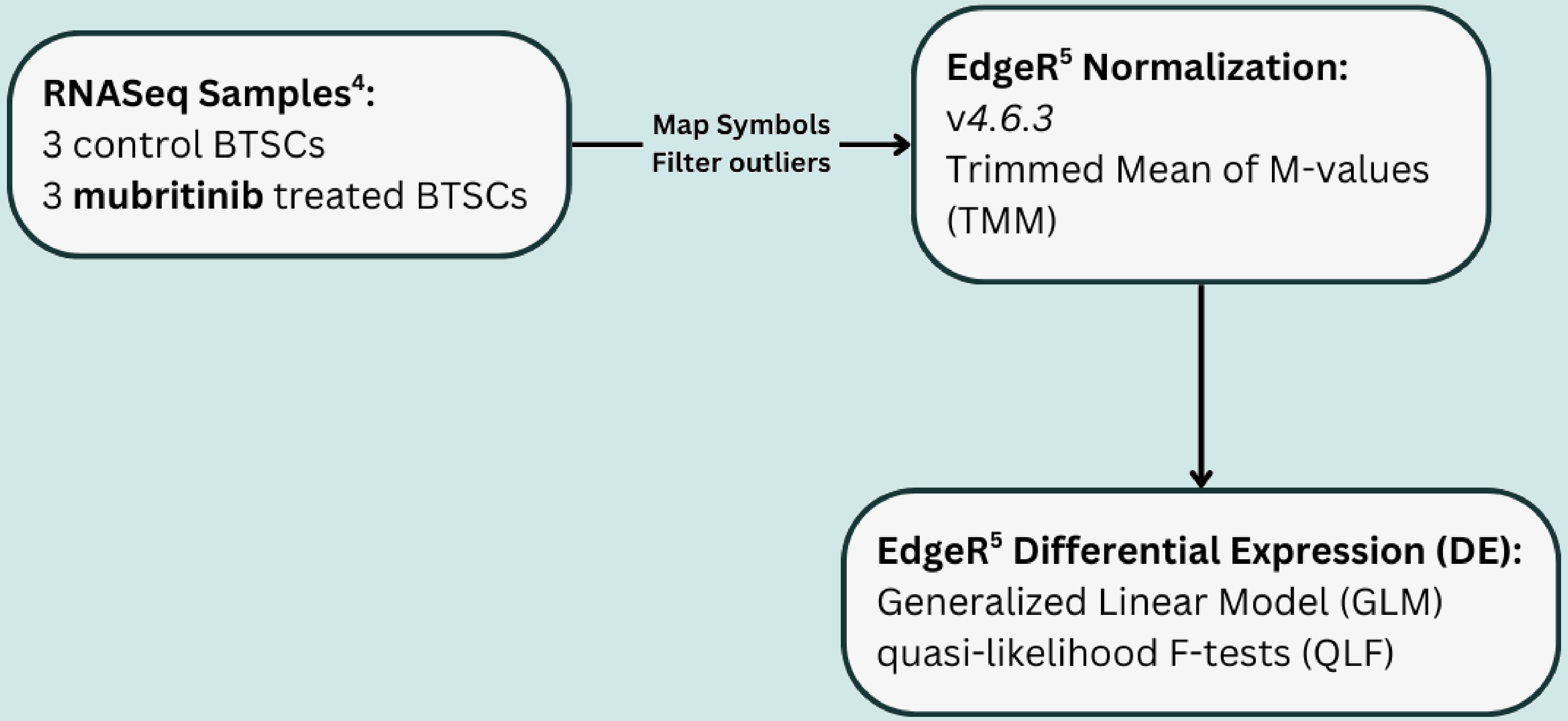


Figure created with biorender:

1. Prager, 2020
2. Agnihotri and Zadeh, 2016
3. Baccelli, 2019

How does Mubritinib treatment affect the biological pathways in BTSCs?

Dataset & Preprocessing



---

4. Burban, 2025
5. Chen, 2025

Dataset & Preprocessing

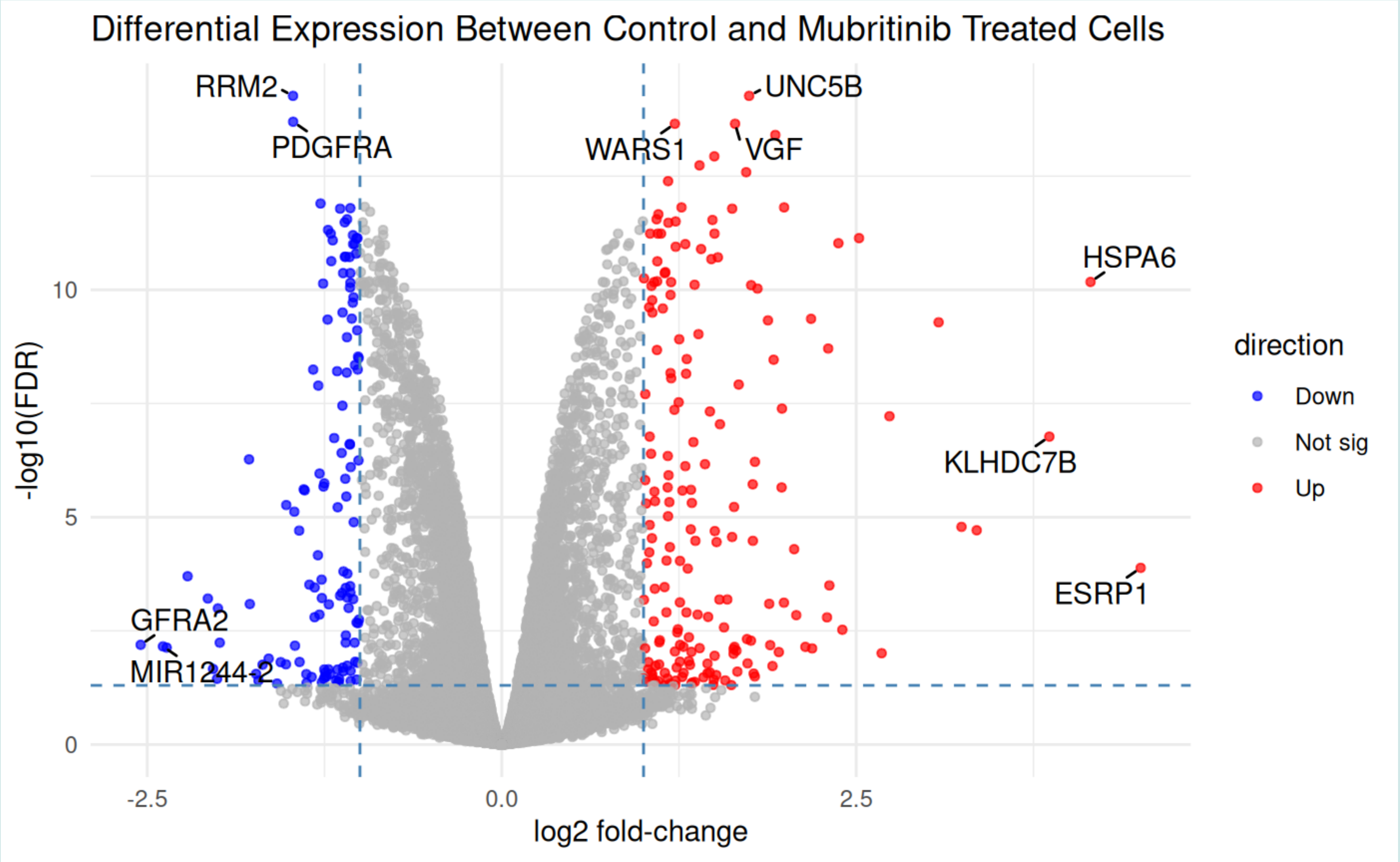
RNASeq Samples[4]:
3 control BTSCs
3 **mubritinib** treated BTSCs

→ **Map Symbols**
Filter outliers →

EdgeR[5] Normalization:
v4.6.3
Trimmed Mean of M-values (TMM)

↓

EdgeR[5] Differential Expression (DE):
Generalized Linear Model (GLM)
quasi-likelihood F-tests (QLF)



Differential expression volcano plot, color thresholded at FDR < 0.05 and an absolute log2 fold-change > 1

4. Burban, 2025
5. Chen, 2025

Methodology: Thresholded

Thresholded gene list:
Top 300 up/down/dis-regulated genes from edgeR DE analysis.

Chosen pathways:

- Gene Ontology Biological Process (GOBP) no IEA *v2025-03*
- Reactome *v2025-05*
- Wikipathways *v2025-05*

gprofiler2[6] - R package (*v0.2.4*)

```
gost_res <- gprofiler2::gost(
  thresholded_genes,
  correction_method = "fdr",
  sources = chosen_pathways)
```

Gene sets then filtered
Max size: 350
Min size: 5

6. Kolberg, 2020

Methodology: Non-thresholded

Gene	logFC
ESRP1	4.506072
HSPA6	4.152318
KLHDC7B	3.861886
MOV10L1	3.349543
...	...
MCIDAS	-2.215471
SNORA9B	-2.363198
MIR1244-2	-2.389393
GFRA2	-2.546541

Ranked Gene List
All genes from edgeR DE analysis

Pathways data (.gmt file)
From the Bader lab[7]
- Sep 1, 2025
- GO BP
- no PFOCR
- no GO IEA

Hallmark

GOBP

Wiki pathways

Reactome

FGSEA[8] - R package (*v1.34.2*)

```
gsea_res <- fgsea::fgsea(
    pathways_data,
    ranked_gene_list,
    maxSize = 350,
    minSize = 5)
```

7. Isserlin, 2014
8. Korotkevich, 2019

Results

Name <chr>	P-value <dbl>
regulation of apoptotic signaling pathway	2.181505e-10
negative regulation of apoptotic signaling pathway	1.154665e-09
tRNA aminoacylation for protein translation	8.973075e-09
amino acid activation	1.690226e-08
tRNA aminoacylation	1.690226e-08
amino acid metabolic process	1.965322e-07
intrinsic apoptotic signaling pathway	7.616440e-07
amino acid import across plasma membrane	2.785634e-06
regulation of intrinsic apoptotic signaling pathway	5.677667e-06
neutral amino acid transport	6.838737e-06

Up-regulated thresholded over-representation analysis. Top ten gene sets are shown.

Name <chr>	P-value <dbl>
DNA replication	4.955923e-53
DNA-templated DNA replication	7.479999e-45
sister chromatid segregation	1.160430e-40
mitotic sister chromatid segregation	7.168424e-40
nuclear chromosome segregation	7.735860e-40
mitotic nuclear division	1.543157e-38
positive regulation of cell cycle process	2.115258e-38
regulation of mitotic cell cycle phase transition	3.413224e-35
positive regulation of cell cycle	1.104583e-34
negative regulation of cell cycle process	5.382157e-34

Down-regulated thresholded over-representation analysis. Top ten gene sets are shown.

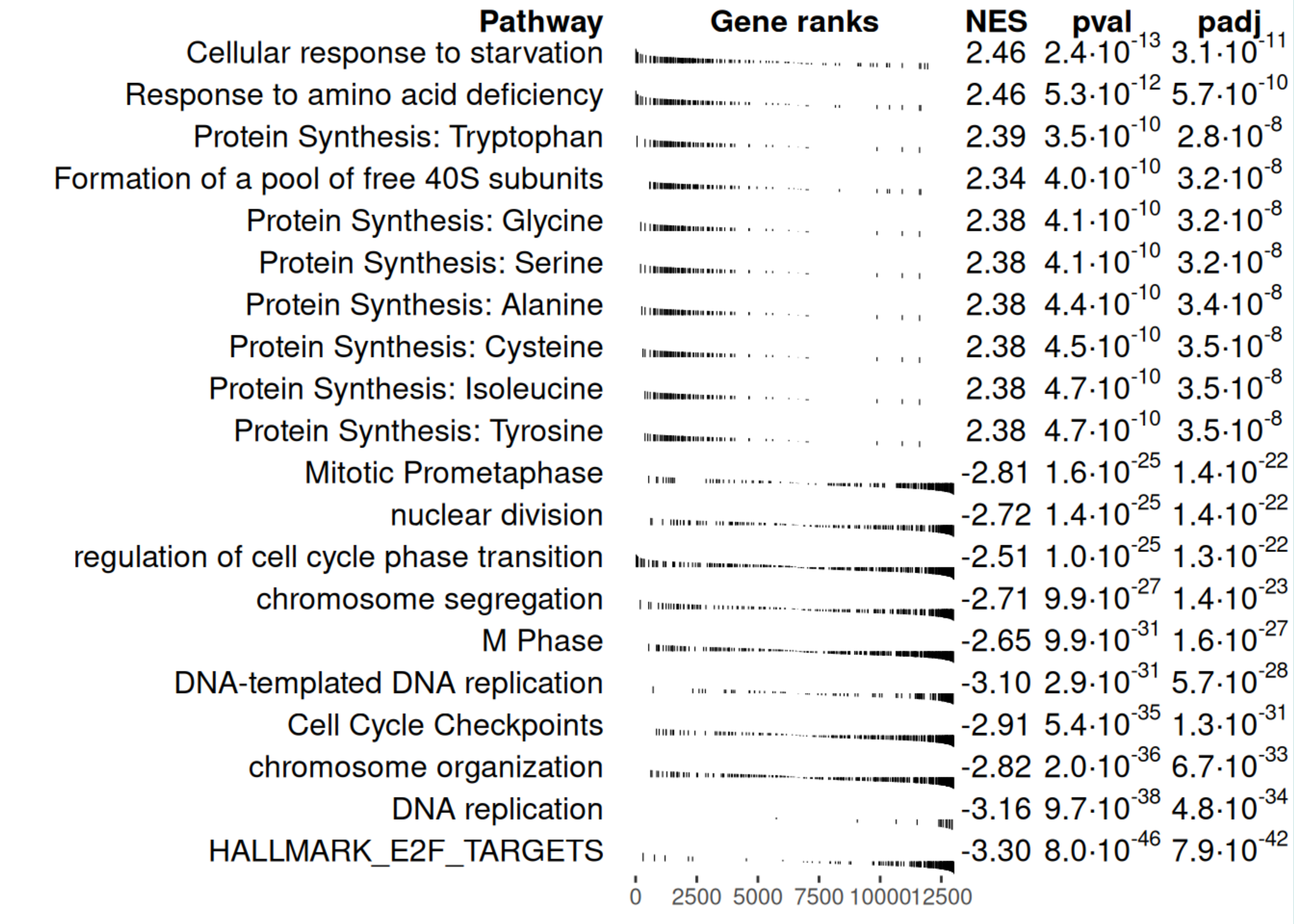
Results

Name <chr>	P-value <dbl>
regulation of apoptotic signaling pathway	2.181505e-10
negative regulation of apoptotic signaling pathway	1.154665e-09
tRNA aminoacylation for protein translation	8.973075e-09
amino acid activation	1.690226e-08
tRNA aminoacylation	1.690226e-08
amino acid metabolic process	1.965322e-07
intrinsic apoptotic signaling pathway	7.616440e-07
amino acid import across plasma membrane	2.785634e-06
regulation of intrinsic apoptotic signaling pathway	5.677667e-06
neutral amino acid transport	6.838737e-06

Up-regulated thresholded over-representation analysis. Top ten gene sets are shown.

Name <chr>	P-value <dbl>
DNA replication	4.955923e-53
DNA-templated DNA replication	7.479999e-45
sister chromatid segregation	1.160430e-40
mitotic sister chromatid segregation	7.168424e-40
nuclear chromosome segregation	7.735860e-40
mitotic nuclear division	1.543157e-38
positive regulation of cell cycle process	2.115258e-38
regulation of mitotic cell cycle phase transition	3.413224e-35
positive regulation of cell cycle	1.104583e-34
negative regulation of cell cycle process	5.382157e-34

Down-regulated thresholded over-representation analysis. Top ten gene sets are shown.



Pathway	NES	pval	padj
Cellular response to starvation	2.46	$2.4 \cdot 10^{-13}$	$3.1 \cdot 10^{-11}$
Response to amino acid deficiency	2.46	$5.3 \cdot 10^{-12}$	$5.7 \cdot 10^{-10}$
Protein Synthesis: Tryptophan	2.39	$3.5 \cdot 10^{-10}$	$2.8 \cdot 10^{-8}$
Formation of a pool of free 40S subunits	2.34	$4.0 \cdot 10^{-10}$	$3.2 \cdot 10^{-8}$
Protein Synthesis: Glycine	2.38	$4.1 \cdot 10^{-10}$	$3.2 \cdot 10^{-8}$
Protein Synthesis: Serine	2.38	$4.1 \cdot 10^{-10}$	$3.2 \cdot 10^{-8}$
Protein Synthesis: Alanine	2.38	$4.4 \cdot 10^{-10}$	$3.4 \cdot 10^{-8}$
Protein Synthesis: Cysteine	2.38	$4.5 \cdot 10^{-10}$	$3.5 \cdot 10^{-8}$
Protein Synthesis: Isoleucine	2.38	$4.7 \cdot 10^{-10}$	$3.5 \cdot 10^{-8}$
Protein Synthesis: Tyrosine	2.38	$4.7 \cdot 10^{-10}$	$3.5 \cdot 10^{-8}$
Mitotic Prometaphase	-2.81	$1.6 \cdot 10^{-25}$	$1.4 \cdot 10^{-22}$
nuclear division	-2.72	$1.4 \cdot 10^{-25}$	$1.4 \cdot 10^{-22}$
regulation of cell cycle phase transition	-2.51	$1.0 \cdot 10^{-25}$	$1.3 \cdot 10^{-22}$
chromosome segregation	-2.71	$9.9 \cdot 10^{-27}$	$1.4 \cdot 10^{-23}$
M Phase	-2.65	$9.9 \cdot 10^{-31}$	$1.6 \cdot 10^{-27}$
DNA-templated DNA replication	-3.10	$2.9 \cdot 10^{-31}$	$5.7 \cdot 10^{-28}$
Cell Cycle Checkpoints	-2.91	$5.4 \cdot 10^{-35}$	$1.3 \cdot 10^{-31}$
chromosome organization	-2.82	$2.0 \cdot 10^{-36}$	$6.7 \cdot 10^{-33}$
DNA replication	-3.16	$9.7 \cdot 10^{-38}$	$4.8 \cdot 10^{-34}$
HALLMARK_E2F_TARGETS	-3.30	$8.0 \cdot 10^{-46}$	$7.9 \cdot 10^{-42}$

Non-thresholded gene set enrichment analysis. Top ten gene sets for both positive and negative classes are shown.

Results

Name <chr>	P-value <dbl>
regulation of apoptotic signaling pathway	2.181505e-10
negative regulation of apoptotic signaling pathway	1.154665e-09
tRNA aminoacylation for protein translation	8.973075e-09
amino acid activation	1.690226e-08
tRNA aminoacylation	1.690226e-08
amino acid metabolic process	1.965322e-07
intrinsic apoptotic signaling pathway	7.616440e-07
amino acid import across plasma membrane	2.785634e-06
regulation of intrinsic apoptotic signaling pathway	5.677667e-06
neutral amino acid transport	6.838737e-06

Up-regulated thresholded over-representation analysis. Top ten gene sets are shown.

Name <chr>	P-value <dbl>
DNA replication	4.955923e-53
DNA-templated DNA replication	7.479999e-45
sister chromatid segregation	1.160430e-40
mitotic sister chromatid segregation	7.168424e-40
nuclear chromosome segregation	7.735860e-40
mitotic nuclear division	1.543157e-38
positive regulation of cell cycle process	2.115258e-38
regulation of mitotic cell cycle phase transition	3.413224e-35
positive regulation of cell cycle	1.104583e-34
negative regulation of cell cycle process	5.382157e-34

Down-regulated thresholded over-representation analysis. Top ten gene sets are shown.

Pathway	Gene ranks	NES	pval	padj
Cellular response to starvation		2.46	$2.4\cdot10^{-13}$	$3.1\cdot10^{-11}$
Response to amino acid deficiency		2.46	$5.3\cdot10^{-12}$	$5.7\cdot10^{-10}$
Protein Synthesis: Tryptophan		2.39	$3.5\cdot10^{-10}$	$2.8\cdot10^{-8}$
Formation of a pool of free 40S subunits		2.34	$4.0\cdot10^{-10}$	$3.2\cdot10^{-8}$
Protein Synthesis: Glycine		2.38	$4.1\cdot10^{-10}$	$3.2\cdot10^{-8}$
Protein Synthesis: Serine		2.38	$4.1\cdot10^{-10}$	$3.2\cdot10^{-8}$
Protein Synthesis: Alanine		2.38	$4.4\cdot10^{-10}$	$3.4\cdot10^{-8}$
Protein Synthesis: Cysteine		2.38	$4.5\cdot10^{-10}$	$3.5\cdot10^{-8}$
Protein Synthesis: Isoleucine		2.38	$4.7\cdot10^{-10}$	$3.5\cdot10^{-8}$
Protein Synthesis: Tyrosine		2.38	$4.7\cdot10^{-10}$	$3.5\cdot10^{-8}$
Mitotic Prometaphase		-2.81	$1.6\cdot10^{-25}$	$1.4\cdot10^{-22}$
nuclear division		-2.72	$1.4\cdot10^{-25}$	$1.4\cdot10^{-22}$
regulation of cell cycle phase transition		-2.51	$1.0\cdot10^{-25}$	$1.3\cdot10^{-22}$
chromosome segregation		-2.71	$9.9\cdot10^{-27}$	$1.4\cdot10^{-23}$
M Phase		-2.65	$9.9\cdot10^{-31}$	$1.6\cdot10^{-27}$
DNA-templated DNA replication		-3.10	$2.9\cdot10^{-31}$	$5.7\cdot10^{-28}$
Cell Cycle Checkpoints		-2.91	$5.4\cdot10^{-35}$	$1.3\cdot10^{-31}$
chromosome organization		-2.82	$2.0\cdot10^{-36}$	$6.7\cdot10^{-33}$
DNA replication		-3.16	$9.7\cdot10^{-38}$	$4.8\cdot10^{-34}$
HALLMARK_E2F_TARGETS		-3.30	$8.0\cdot10^{-46}$	$7.9\cdot10^{-42}$
	0 2500 5000 7500 10000 12500			

Non-thresholded gene set enrichment analysis. Top ten gene sets for both positive and negative classes are shown.

Results

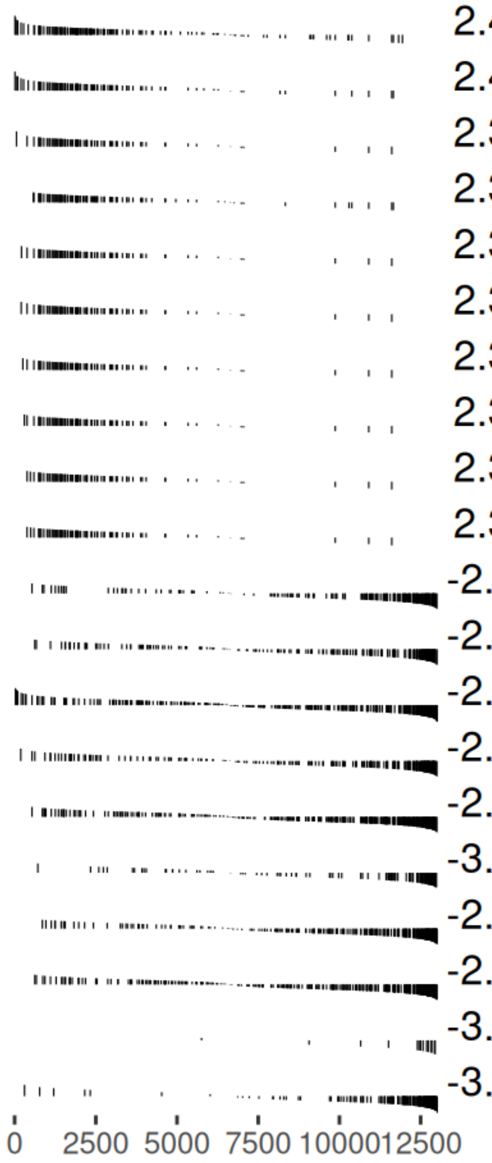
Name <chr>	P-value <dbl>
regulation of apoptotic signaling pathway	2.181505e-10
negative regulation of apoptotic signaling pathway	1.154665e-09
tRNA aminoacylation for protein translation	8.973075e-09
amino acid activation	1.690226e-08
tRNA aminoacylation	1.690226e-08
amino acid metabolic process	1.965322e-07
intrinsic apoptotic signaling pathway	7.616440e-07
amino acid import across plasma membrane	2.785634e-06
regulation of intrinsic apoptotic signaling pathway	5.677667e-06
neutral amino acid transport	6.838737e-06

Up-regulated thresholded over-representation analysis. Top ten gene sets are shown.

Name <chr>	P-value <dbl>
DNA replication	4.955923e-53
DNA-templated DNA replication	7.479999e-45
sister chromatid segregation	1.160430e-40
mitotic sister chromatid segregation	7.168424e-40
nuclear chromosome segregation	7.735860e-40
mitotic nuclear division	1.543157e-38
positive regulation of cell cycle process	2.115258e-38
regulation of mitotic cell cycle phase transition	3.413224e-35
positive regulation of cell cycle	1.104583e-34
negative regulation of cell cycle process	5.382157e-34

Down-regulated thresholded over-representation analysis. Top ten gene sets are shown.

Pathway	Gene ranks	NES	pval	padj
Cellular response to starvation		2.46	$2.4\cdot10^{-13}$	$3.1\cdot10^{-11}$
Response to amino acid deficiency		2.46	$5.3\cdot10^{-12}$	$5.7\cdot10^{-10}$
Protein Synthesis: Tryptophan		2.39	$3.5\cdot10^{-10}$	$2.8\cdot10^{-8}$
Formation of a pool of free 40S subunits		2.34	$4.0\cdot10^{-10}$	$3.2\cdot10^{-8}$
Protein Synthesis: Glycine		2.38	$4.1\cdot10^{-10}$	$3.2\cdot10^{-8}$
Protein Synthesis: Serine		2.38	$4.1\cdot10^{-10}$	$3.2\cdot10^{-8}$
Protein Synthesis: Alanine		2.38	$4.4\cdot10^{-10}$	$3.4\cdot10^{-8}$
Protein Synthesis: Cysteine		2.38	$4.5\cdot10^{-10}$	$3.5\cdot10^{-8}$
Protein Synthesis: Isoleucine		2.38	$4.7\cdot10^{-10}$	$3.5\cdot10^{-8}$
Protein Synthesis: Tyrosine		2.38	$4.7\cdot10^{-10}$	$3.5\cdot10^{-8}$
Mitotic Prometaphase		-2.81	$1.6\cdot10^{-25}$	$1.4\cdot10^{-22}$
nuclear division		-2.72	$1.4\cdot10^{-25}$	$1.4\cdot10^{-22}$
regulation of cell cycle phase transition		-2.51	$1.0\cdot10^{-25}$	$1.3\cdot10^{-22}$
chromosome segregation		-2.71	$9.9\cdot10^{-27}$	$1.4\cdot10^{-23}$
M Phase		-2.65	$9.9\cdot10^{-31}$	$1.6\cdot10^{-27}$
DNA-templated DNA replication		-3.10	$2.9\cdot10^{-31}$	$5.7\cdot10^{-28}$
Cell Cycle Checkpoints		-2.91	$5.4\cdot10^{-35}$	$1.3\cdot10^{-31}$
chromosome organization		-2.82	$2.0\cdot10^{-36}$	$6.7\cdot10^{-33}$
DNA replication		-3.16	$9.7\cdot10^{-38}$	$4.8\cdot10^{-34}$
HALLMARK_E2F_TARGETS		-3.30	$8.0\cdot10^{-46}$	$7.9\cdot10^{-42}$



Non-thresholded gene set enrichment analysis. Top ten gene sets for both positive and negative classes are shown.

Results

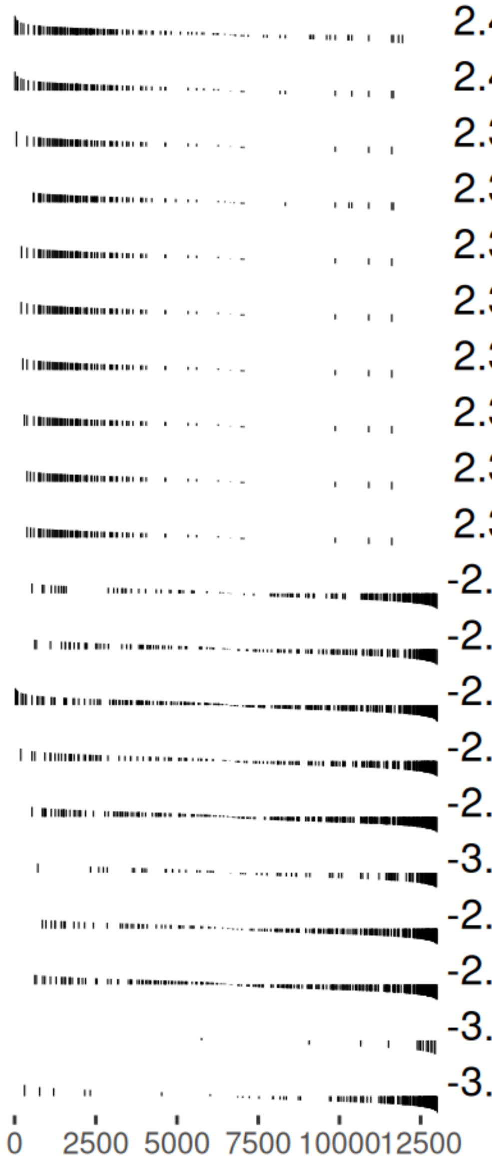
Name <chr>	P-value <dbl>
regulation of apoptotic signaling pathway	2.181505e-10
negative regulation of apoptotic signaling pathway	1.154665e-09
tRNA aminoacylation for protein translation	8.973075e-09
amino acid activation	1.690226e-08
tRNA aminoacylation	1.690226e-08
amino acid metabolic process	1.965322e-07
intrinsic apoptotic signaling pathway	7.616440e-07
amino acid import across plasma membrane	2.785634e-06
regulation of intrinsic apoptotic signaling pathway	5.677667e-06
neutral amino acid transport	6.838737e-06

Up-regulated thresholded over-representation analysis. Top ten gene sets are shown.

Name <chr>	P-value <dbl>
DNA replication	4.955923e-53
DNA-templated DNA replication	7.479999e-45
sister chromatid segregation	1.160430e-40
mitotic sister chromatid segregation	7.168424e-40
nuclear chromosome segregation	7.735860e-40
mitotic nuclear division	1.543157e-38
positive regulation of cell cycle process	2.115258e-38
regulation of mitotic cell cycle phase transition	3.413224e-35
positive regulation of cell cycle	1.104583e-34
negative regulation of cell cycle process	5.382157e-34

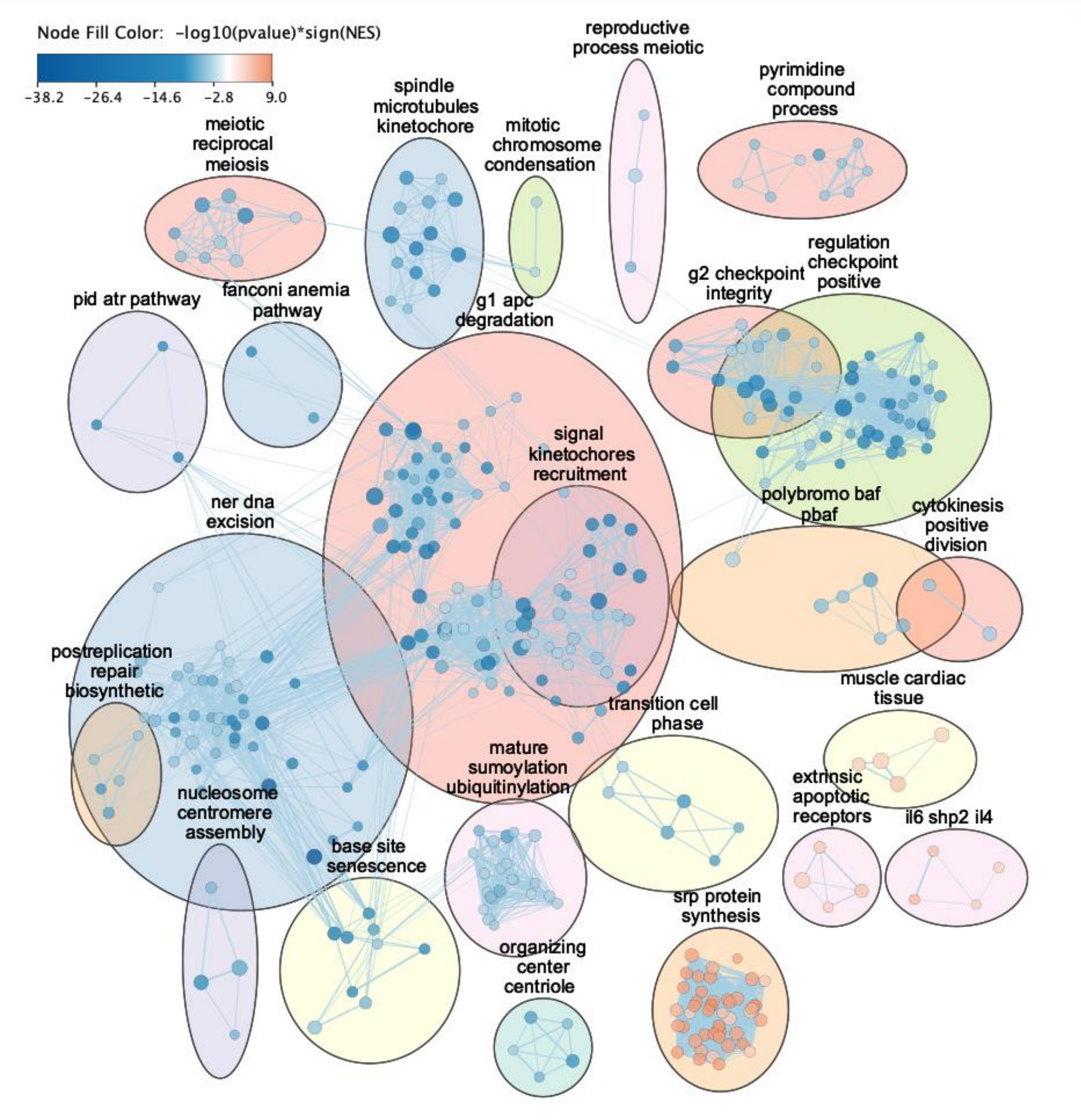
Down-regulated thresholded over-representation analysis. Top ten gene sets are shown.

Pathway	Gene ranks	NES	pval	padj
Cellular response to starvation		2.46	$2.4\cdot10^{-13}$	$3.1\cdot10^{-11}$
Response to amino acid deficiency		2.46	$5.3\cdot10^{-12}$	$5.7\cdot10^{-10}$
Protein Synthesis: Tryptophan		2.39	$3.5\cdot10^{-10}$	$2.8\cdot10^{-8}$
Formation of a pool of free 40S subunits		2.34	$4.0\cdot10^{-10}$	$3.2\cdot10^{-8}$
Protein Synthesis: Glycine		2.38	$4.1\cdot10^{-10}$	$3.2\cdot10^{-8}$
Protein Synthesis: Serine		2.38	$4.1\cdot10^{-10}$	$3.2\cdot10^{-8}$
Protein Synthesis: Alanine		2.38	$4.4\cdot10^{-10}$	$3.4\cdot10^{-8}$
Protein Synthesis: Cysteine		2.38	$4.5\cdot10^{-10}$	$3.5\cdot10^{-8}$
Protein Synthesis: Isoleucine		2.38	$4.7\cdot10^{-10}$	$3.5\cdot10^{-8}$
Protein Synthesis: Tyrosine		2.38	$4.7\cdot10^{-10}$	$3.5\cdot10^{-8}$
Mitotic Prometaphase		-2.81	$1.6\cdot10^{-25}$	$1.4\cdot10^{-22}$
nuclear division		-2.72	$1.4\cdot10^{-25}$	$1.4\cdot10^{-22}$
regulation of cell cycle phase transition		-2.51	$1.0\cdot10^{-25}$	$1.3\cdot10^{-22}$
chromosome segregation		-2.71	$9.9\cdot10^{-27}$	$1.4\cdot10^{-23}$
M Phase		-2.65	$9.9\cdot10^{-31}$	$1.6\cdot10^{-27}$
DNA-templated DNA replication		-3.10	$2.9\cdot10^{-31}$	$5.7\cdot10^{-28}$
Cell Cycle Checkpoints		-2.91	$5.4\cdot10^{-35}$	$1.3\cdot10^{-31}$
chromosome organization		-2.82	$2.0\cdot10^{-36}$	$6.7\cdot10^{-33}$
DNA replication		-3.16	$9.7\cdot10^{-38}$	$4.8\cdot10^{-34}$
HALLMARK_E2F_TARGETS		-3.30	$8.0\cdot10^{-46}$	$7.9\cdot10^{-42}$



Non-thresholded gene set enrichment analysis. Top ten gene sets for both positive and negative classes are shown.

Results

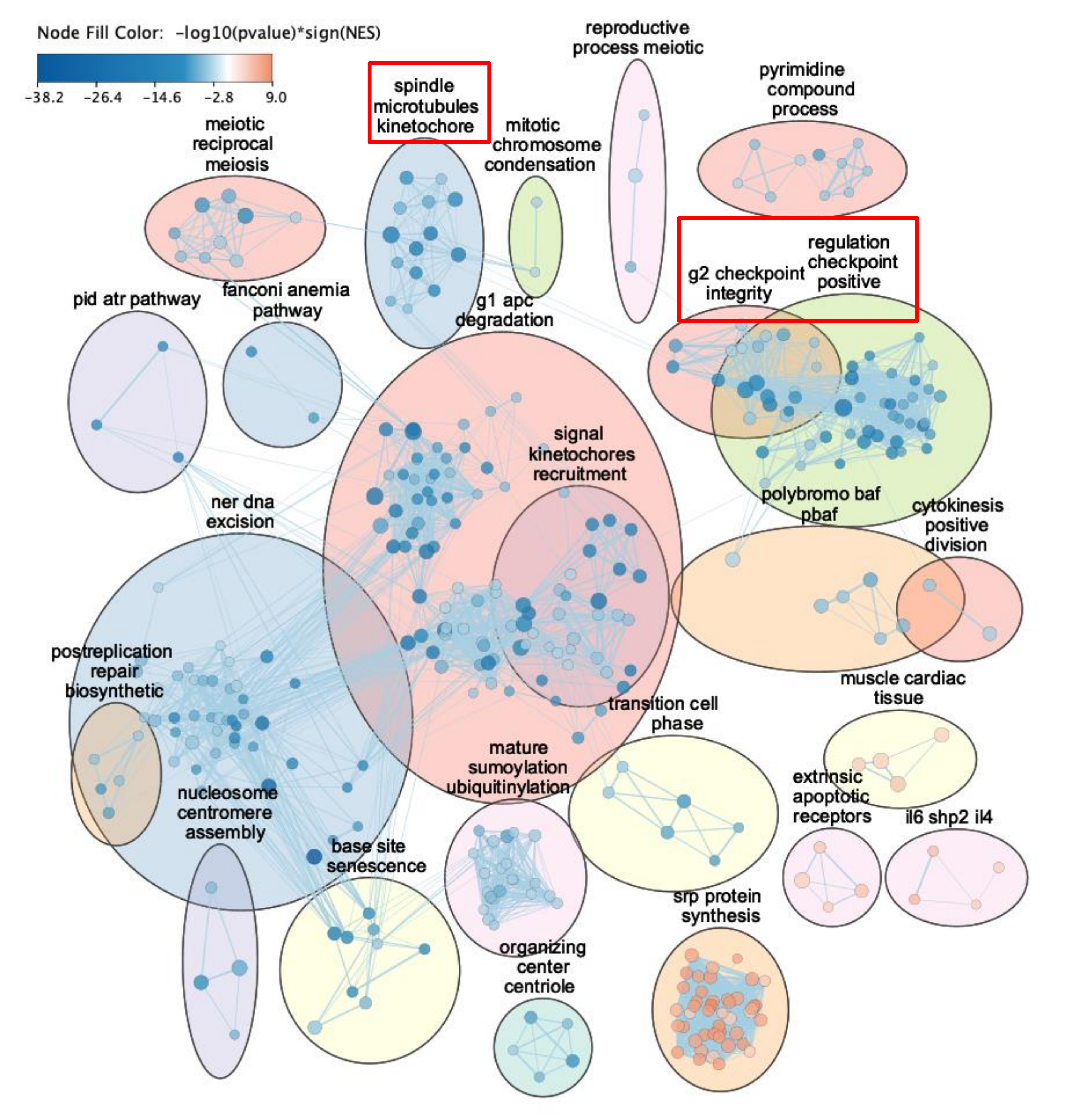


Annotated and curated enrichment map network for non-thresholded analysis, using Cytoscape. Created with 0.005 Q-value and a 0.375 edge similarity cutoff, annotated with AutoAnnotate.

Results

Down-regulation of the cell cycle:

1. Cancer cells are likely to over-engage the cell cycle.[9]
2. Metabolic stress causes cell cycle arrest.[10]



Annotated and curated enrichment map network for non-thresholded analysis, using Cytoscape. Created with 0.005 Q-value and a 0.375 edge similarity cutoff, annotated with AutoAnnotate.

9. Shapiro, 1999
10. Tuo, 2019

Results

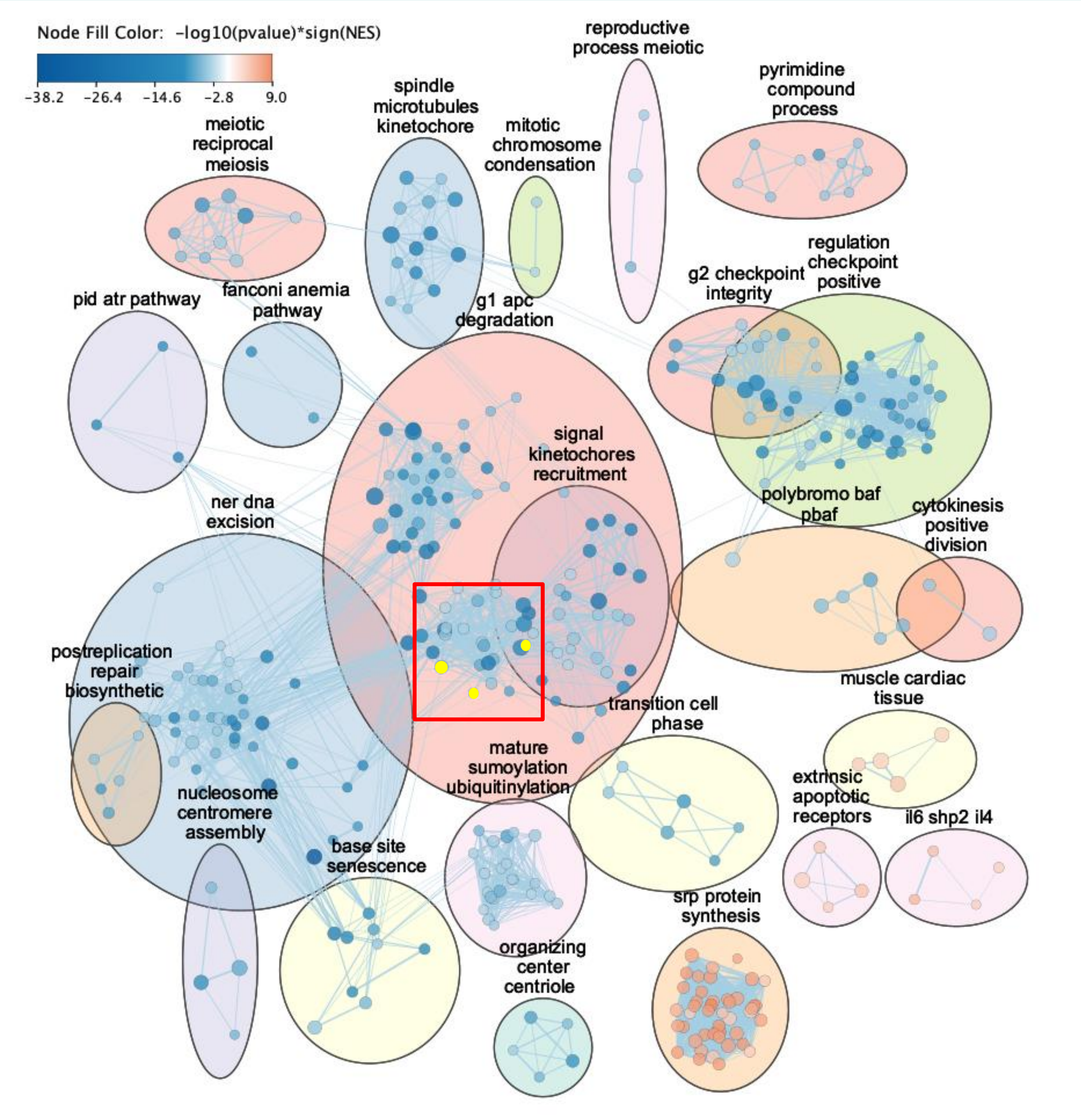
Down-regulation of the cell cycle:

1. Cancer cells are likely to over-engage the cell cycle.[9]
2. Metabolic stress causes cell cycle arrest.[10]

Down-regulation of AMPK/P27 signaling related pathways:

- cyclin D1-associated events in G1
- SCF SKP2-mediated degradation of p27/p21
- E2F-mediated regulation of DNA replication

As described in Burban et al.[4]



Annotated and curated enrichment map network for non-thresholded analysis, using Cytoscape. Created with 0.005 Q-value and a 0.375 edge similarity cutoff, annotated with AutoAnnotate.

4. Buban, 2025
9. Shapiro, 1999
10. Tuo, 2019

Results

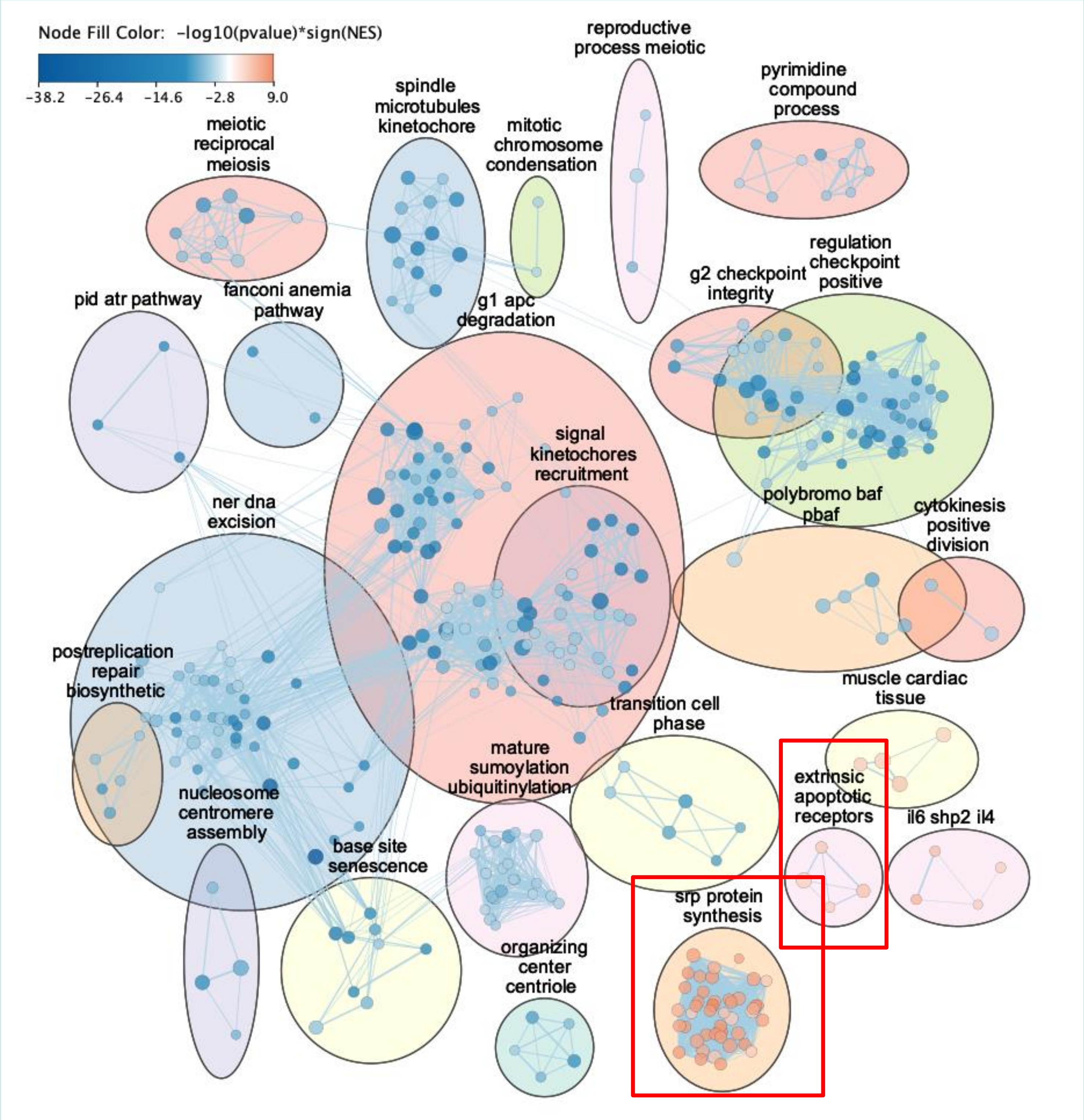
Down-regulation of the cell cycle:

1. Cancer cells are likely to over-engage the cell cycle.[9]
2. Metabolic stress causes cell cycle arrest.[10]

Down-regulation of AMPK/P27 signaling related pathways:

- cyclin D1-associated events in G1
- SCF SKP2-mediated degradation of p27/p21
- E2F-mediated regulation of DNA replication

As described in Burban et al.[4]



Annotated and curated enrichment map network for non-thresholded analysis, using Cytoscape. Created with 0.005 Q-value and a 0.375 edge similarity cutoff, annotated with AutoAnnotate.

Up-regulation of apoptosis:

- Metabolic stress can cause apoptosis signaling.[11]

Up-regulation of protein synthesis

- Metabolic stress can cause metabolic rewiring.[12]

4. Burban, 2025
9. Shapiro, 1999
10. Tuo, 2019
11. El-Osta, 2016
12. Brillo, 2021

Mubritinib

Previously known to affect mitochondrial oxidative phosphorylation.[3]

— Treated to BTSCs →

Loss of energy causes massive down-regulation of **cell cycle** and up-regulation of **apoptosis.**

— As a therapeutic →

Reduce viability of cancer stem cells!

3. Baccelli, 2019

References

1. Prager, B. C., Bhargava, S., Mahadev, V., Hubert, C. G., & Rich, J. N. (2020). Glioblastoma stem cells: Driving resilience through chaos. *Trends in Cancer*, *6*(3), 223–235. doi:10.1016/j.trecan.2020.01.009
2. Agnihotri, S., & Zadeh, G. (2016). Metabolic reprogramming in glioblastoma: the influence of cancer metabolism on epigenetics and unanswered questions. *Neuro-Oncology*, *18*(2), 160–172. doi:10.1093/neuonc/nov125
3. Burban, A., Tessier, C., Larroquette, M., Guyon, J., Lubiato, C., Pinglaut, M., … Sharanek, A. (2025). Exploiting metabolic vulnerability in glioblastoma using a brain-penetrant drug with a safe profile. *EMBO Molecular Medicine*, *17*(3), 469–503. doi:10.1038/s44321-025-00195-6
4. Baccelli, I., Gareau, Y., Lehnertz, B., Gingras, S., Spinella, J.-F., Corneau, S., … Sauvageau, G. (2019). Mubritinib targets the electron transport chain complex I and reveals the landscape of OXPHOS dependency in acute myeloid leukemia. *Cancer Cell*, *36*(1), 84-99.e8. doi:10.1016/j.ccell.2019.06.003
5. Chen, Y., Chen, L., Lun, A. T. L., Baldoni, P., & Smyth, G. K. (2025). edgeR v4: Powerful differential analysis of sequencing data with expanded functionality and improved support for small counts and larger datasets. *Nucleic Acids Research, 53*(2), gkaf018. https://doi.org/10.1093/nar/gkaf018
6. Kolberg, L., Raudvere, U., Kuzmin, I., Vilo, J., & Peterson, H. (2020). gprofiler2—An R package for gene list functional enrichment analysis and namespace conversion toolset g:Profiler. *F1000Research, 9*, 709. https://doi.org/10.12688/f1000research.24956.1
7. Isserlin, R., Merico, D., Voisin, V., & Bader, G. D. (2014). Enrichment map: A Cytoscape app to visualize and explore OMICs pathway enrichment results. *F1000Research, 3*, 141. doi:10.12688/f1000research.4536.1
8. Korotkevich, G., Sukhov, V., & Sergushichev, A. (2019). Fast gene set enrichment analysis. *bioRxiv*. https://doi.org/10.1101/060012
9. Shapiro, G. I., & Harper, J. W. (1999). Anticancer drug targets: Cell cycle and checkpoint control. *Journal of Clinical Investigation, 104*(12), 1645–1653. https://doi.org/10.1172/JCI9054
10. Tuo, L., Xiang, J., Pan, X., et al. (2019). PCK1 negatively regulates cell cycle progression and hepatoma cell proliferation via the AMPK/p27Kip1 axis. *Journal of Experimental & Clinical Cancer Research, 38*(1), 50. https://doi.org/10.1186/s13046-019-1029-y
11. El-Osta, H., & Circu, M. L. (2016). Mitochondrial ROS and apoptosis. In L. Buhlman (Ed.), *Mitochondrial mechanisms of degeneration and repair in Parkinson's disease* (pp. 1–23). Springer. https://doi.org/10.1007/978-3-319-42139-1_1
12. Brillo, V., Chieregato, L., Leanza, L., Muccioli, S., & Costa, R. (2021). Mitochondrial dynamics, ROS, and cell signaling: A blended overview. *Life, 11*(4), 332. https://doi.org/10.3390/life11040332

Questions?